Appendix A

The following table is the complete packaging machinery equipment tree from www.packexpo.com, the tradeshow website of the Packaging Machinery Manufacturers Institute (PMMI).

- Packaging Machinery
  - Adhesive/Glue Applicators
  - Cartoning Machines
    - Carton Blank Erecting
    - Carton Closing Machines
    - End Flap Carton Closing
    - Horizontal End Load Cartoner
    - Mandrel Bag-In-Box
    - Mandrel Carton Form, Fill & Seal
    - Skillet Erecting
    - Three Flap Carton Closing
    - Top Load Carton Form, Fill & Seal
    - Tray Erect Load & Seal
    - Vertical Cartoner
    - Wrap-Around Cartoner
    - Wrap-Around Sleeving
  - Case/Tray Erecting Machines
    - Case Erecting
    - Division Inserting

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- Tray Erecting
- Case/Tray Handling Machines
  - Crate Stacking/Unstacking
  - Tray Denesting
  - Tray Stacker
- Case/Tray Loading/Unloading Machines
  - Drop Packing
  - Horizontal Packing
  - Place Packing/Unpacking
- Case/Tray Sealing Machines
  - Case Gluing
  - Case Stapling
  - Case Tapers
  - Wraparound Lidding
- Case/Tray Wrapping Machines
  - Banding
  - Bundling
  - Case Forming & Closing
  - Case Handling Equipment
  - Case Loaders For Bags
  - Case Loaders For Bottles/Jars
  - Case Loaders For Cartons
  - Case Loading
  - Case Opening
  - Case Packing
  - Multipacking Equipment
  - Multipacks For Aseptic Packaging
  - Unit Packaging
- Cleaning Machines
  - Air Cleaning
  - Bottle Washing
  - Crate Washing
  - Dust Collect/Remove
  - Rinsing
- Closing Machines
  - Aerosol Valve Closing
  - Can Seaming
  - Caps, Bottle/Easy Open
- Caps, Bottle/Tamperproof
  - Clip Closing
  - Cork Wiring
  - Crimp Closing

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- Crown Capping
- Fitment Applicator
- Foil Sealing
- Fold Closing
- Fusion Sealing
- Glue Sealing
- Gummed Tape Sealing
- Heat Sealing
- Induction Sealing
- Nail Closing
- Overcapping
- Plugging; Corking
- Press-On Lidding
- Pump Applicator
- Rivet Closing
- Roll-On Capping
- Screw Capping
- Sewing
- Solder Sealing
- Sonic/Ultrasonic and RF Sealing
- Spin Welding
- Staple Closing
- Strapping
- Tamper Evident Seals & Bands
- Tape Sealing
- Tuck Closing
- Twist-Tie Closing
- Weld Sealing

Container & Component Handling Machines

- Accumulating
- Accumulating & Collating
- Accumulation Table
- Arranging Machines/Feeding, Orienting
- Bag Handling
- Bag Presenting
- Barrel & Drum Handling
- Bottle Handling
- Box Handling Systems
- Can Handling
- Carton Handling
- Case Handling
- Component Unscrambler

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- Conveying Machines
- Dispensing Machines
- Handle Applicator
- Inserting
- Key Applicator
- Layer Pad Dispenser
- Leaflet Feeder
- Pallet Dispenser
- Pouch Handling
- Puck Handling, Flow Assistance
- Pucking/Depucking
- Rigid Container Denester
- Rigid Container Orienter
- Rigid Container Single Liner
- Rigid Container Unscrambler
- Robotics
- Sack Seal & Present
- Spoon Applicator
- Straw Applicator
- Tear Tape Applicator
- Top Sheet Dispensing

- Conveyors
  - Air Cushion Conveyor
  - Air Driven Conveyor
  - Apron/Slat Conveyor
  - Bag Conveyor
  - Belt Conveyor
  - Bucket Conveyor
  - Bulk Conveyor
  - Cable Conveyor
  - Chain Conveyor
  - Converging/Dividing Conveyor
  - Drag Chain Conveyor
  - Drum Conveyor
  - Elevating Conveyor
  - Magnetic Conveyor
  - Mobile Conveyor
  - Monorail Conveyor
  - Pallet-Load Conveyor
  - Pneumatic Conveyor
  - Roller Conveyor
  - Screw Conveyor

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- Spiral Conveyor
- Steel Belt Conveyor
- Unit & Package Handling
- Vibratory Conveyor
- Wheel Conveyor
- Wire Mesh Conveyor

- Cooling, Warming & Drying Machines
  - Cooling
  - Drying
  - Warming
- Decorating Machines
  - Capsuling
  - Foiling
  - Shrink/Stretch Sleeving
  - Tag Labeling
- Fill & Seal Machines
  - Ampoule/Vial Fill & Close
  - Bag Fill & Seal
  - Blister Fill & Seal
  - Bottle Fill & Cap
  - Can Fill & Seam
  - Cask Or Keg Fill & Seal
  - Cup/Tub Fill & Seal
  - Fill and Seal Machines Unspecified
  - Sack Fill & Close
  - Tube Fill & Seal
- Filling Machines
  - Aerosol Filler
  - Auger Filler
  - Bag Filler
  - Bag-In-Box Filler
  - Bottle/Jar Filler
  - Can Filler
  - Canning Filler
  - Capsule & Tablet Packaging Filler
  - Carton Filler
  - Case Filler
  - Count Filler
  - Cup Filler
  - Displacement Filler
  - Drum Filler
  - Dry/Solid Product Filler

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- Feeder
- Filling & Closing
- Flow Meter Filler
- Gas & Vacuum Combo Filler
- Gas Flush Filler
- Gravity Filler
- Gross Weighing
- Liquid Filler
- Net Weighing
- Portion Packaging Filler
- Pressure Filler
- Rotating Chamber Filler
- Sterile Filler
- Timed Flow Filler
- Vacuum Filler
- Vacuum Level Filler
- Viscous/Semi-Viscous Filler
- Volumetric Cup Filler
- Volumetric Piston Filler

Form, Fill & Seal Machines

- Blow Mold Fill & Seal Machines
- Carton Form, Fill & Seal Machines
- Deep Draw Form, Fill & Seal, Cold
- Deep Draw Form, Fill & Seal, Thermoform
- Deep Draw Form, Fill & Seal Machines
- Group Package Form, Fill & Seal Machines
- Horizontal Edge Sealing
- Horizontal Flowrapping
- Horizontal Form, Fill & Seal Machines
- Horizontal Lower Reel Flowrapping
- Horizontal Sachet Form, Fill & Seal
- Mandrel Flexible Package Form, Fill & Seal Machines
- Tubular Bag Form, Fill & Seal Machines
- Vertical Form, Fill & Seal Machine For Cartons
- Vertical Form, Fill & Seal Machines
- Vertical Form, Fill & Seal Machines for Pillow Bags or Stand-up
- Vertical Sachet Form, Fill & Seal
- Vertical Strip Packing

Group Packaging Form, Fill & Seal Machines

- Bottom Load Casepacking
- End Load Casepacking
- Plastic Ringing

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- Top Load Casepacking
- Wraparound Casepacking
- Wraparound Sleeving
- Wraparound Traypacking

- Inspecting Machines
  - Aerosol Testing
  - Cap Inspecting
  - Case Inspectors
  - Checkweigher
  - Color Matching
  - Empty Bottle Inspection
  - Fill Height Inspection
  - Foreign Body Detecting
  - Label Inspecting
  - Leak Detectors
  - Open Flap Detector
  - Seal Checking
  - Torque/Tightness Testing
- Labeling Machines
  - Heat Seal Labeling
  - Hot Melt Glue Labeling
  - Pre-Gummed Label Applicator
  - Pressure Sensitive Labeling
  - Print & Apply Labeling
  - Tamper Evident Seals & Bands
  - Weigh Price Labeling
  - Wet Glue Labeling
- Marking Machines
  - Bar Code
  - Coders, Daters & Imprinters
  - Emboss Coder
  - Hot Foil Coder
  - Ink Jet Coder
  - Laser Coder
  - Solid Ink Coder
  - Wet Ink Coder
- Packaging Support/Specialty Equipment
  - Adhesive Equipment
  - Aerosol Equipment
  - Ampule
  - Aseptic Packaging Systems
  - Assembling

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- Bag Compression
- Blister Equipment
- Bottle Uncasers
- Card Packaging
- Cartridge Machinery
- Count/Cotton/Stack
- Foam In Place
- Hooding
- Identifying
- Lab/Pilot Plant
- Label Stripping
- Lid Banders
- Lidding - Non-Capping
- Material Handling
- Measuring
- Netting Machinery
- Protective Packaging
- Rework/Reverse Operation
- Spraying
- Stacking
- Tablet & Capsule Packaging
- Tea Bag
- Tying
- Vacuum Packaging
- Vial Packaging
- Weigh/Count/Print
- Weighing Systems

- Pallet Forming, Dismantling & Securing Machines
  - Bag Palletizers
  - Can Stacking
  - High Level Palletizer
  - Layer Pad Dispenser
  - Layer Pad Removing
  - Lift-Off Depalletizer
  - Lift-Off Palletizer
  - Low Level Palletizer
  - Multi-Position Palletizer
  - Pallet Dispenser
  - Pallet Inverter
  - Palletize/Depalletize
  - Palletize/Unitize
  - Pallet Shrink-Wrapping

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